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Orbiting Data Centers

They don't yet make business sense, but space and satellites are an increasingly important part of digital transformation and US-China competition.

By James Lewis

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The very successful SpaceX and the somewhat less successful xAI have announced plans to merge and create on-orbit data centers. This is probably some financial machination connected to SpaceX's upcoming IPO. Here's why:

Cost to orbit: Some estimates suggest that the IT equipment for a small-to-medium-sized data center might weigh around 2000 tons, so it could cost SpaceX at least $200 million and require multiple launches to put this in orbit. Dividing the load among many smaller data satellites could make the cost more manageable, but not yet competitive with terrestrial facilities. As SpaceX introduces more powerful rockets or if it can use a "mesh" network approach using multiple small satellites, the cost could drop, but launch costs are now high and not the whole story on feasibility.

Maintenance: Someday soon, there will be robotic repair capabilities for spacecraft, but right now, if something breaks or needs to be replaced, a satellite is out of luck. Repairing the recent outage in a Northern Virginia data center required a manual intervention not possible in space. SpaceX hopes to avoid the problem by using multiple cheap satellites and replacing them as they fail or become obsolete (hardware becomes obsolete in 2-3 years for terrestrial centers).

Energy: Solar energy in space is plentiful, but would require huge solar arrays to collect enough power, adding to the cost of putting a data center in orbit. If lift to orbit becomes cheaper, it's doable, but until then, it's an obstacle.

Cooling: the average temperature in space is less than -400°F, but cooling the servers (done more efficiently with water or air conditioning at terrestrial data centers) would require huge radiators or other heat-transfer systems, adding to weight and cost.

Data transmission: locating a data center in space would probably add a few milliseconds to send or receive traffic, so some commercial uses (finance, gaming) are unfeasible.

Environment: Locating space avoids NIMBY and the need for local, state, and most federal permitting. That's a plus. But space is a hostile environment with debris and, more importantly, radiation, posing risks to satellite operations.

An orbiting data center could service inter-satellite communications (processing remote sensing data on-orbit, for example), but any cost advantage is minimal given the current level of commercial demand. It's likely that someday, locating data processing in space will make commercial sense as the satellite infrastructure continues to expand in numbers and capabilities. Cost-to-orbit is the key variable.

If we are going to go all science-fiction on the topic, how about a data center on the Moon? It's likely that there will be a human presence (either American or Chinese) on the Moon within a decade. A data center will not be a first priority for a lunar presence, and will face the same transmission delay problems as orbiting data

centers, but it will eventually make sense, given the cloud services currently required for robotics, and robots will become a big presence on the Moon. Get used to the idea of lunar infrastructure.

Sounds far-fetched, but we are only 25 years behind schedule.

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Photo: A Falcon 9 rocket launches from SLC-40 at Cape Canaveral Space Force Station, Fla., Jan. 31, 2022. Credit: U.S. Space Force photo by Joshua Conti via Wikimedia Commons.